

POSITION STATEMENT

Position statement: LEPROSY: Diagnosis, treatment and follow-up

Abstract

Background: Leprosy, or Hansen's disease, is a chronic infection caused by *Mycobacterium leprae* (M. leprae) or *Mycobacterium lepromatosis* (M. lepromatosis). In Europe, most leprosy cases are imported, although occasional cases arise from old endemic foci. The disease is often not suspected due to its diminished emphasis in medical education. In the late 20th century, global health priorities shifted toward tuberculosis and HIV, leading to a reduction in the WHO leprosy programme under the assumption that leprosy was nearly eliminated. This lack of awareness contributes to harmful diagnostic delays.

Introduction

Leprosy, also known as Hansen's disease, is a chronic infection caused by *Mycobacterium leprae* or *Mycobacterium lepromatosis*. While most cases in Europe are imported, sporadic cases continue to emerge from historical endemic areas. The condition is frequently overlooked because it is no longer a focus in medical training. During the late twentieth century, attention shifted to tuberculosis (TB) and human immunodeficiency virus (HIV), resulting in a scaling down of the WHO leprosy programme based on the belief that leprosy had been largely eradicated.

In 2005, the WHO declared leprosy eliminated as a public health problem globally. However, this declaration was premature. To date, approximately 250,000 new cases are reported annually worldwide, with stable incidence and prevalence rates. With increasing migration from endemic regions, the risk of leprosy in Europe is rising. Current immigrant populations often live in

overcrowded and suboptimal conditions—factors that may facilitate transmission, unlike earlier migration patterns.

Patients are now being diagnosed across the Western world, often after significant diagnostic delays, leading to lifelong disabilities. Medical curricula provide minimal coverage of leprosy, leaving even specialists such as dermatologists, neurologists, and rheumatologists poorly equipped to recognize it. Lack of awareness remains a primary reason for missed diagnoses. This position statement aims to improve physician awareness and understanding of leprosy in the European context.

Diagnosis

Leprosy primarily affects the peripheral nerves, skin, and mucosa—especially the upper respiratory tract. Skin lesions are typically the first noticeable sign, contributing to its classification as a dermatological condition. If untreated, leprosy can lead to permanent damage to the skin, nerves, limbs, and eyes. Tissue damage results partly from direct *M. leprae* infiltration, but the majority of pathology arises from immunological reactions.

Secondary infections due to trauma—following sensory loss—can lead to tissue destruction. Bone and cartilage resorption may occur, resulting in shortening and deformities of fingers, toes, and the nose (clinical features illustrated in deformed hands due to nerve damage and secondary infection).

Patients may report:

- Loss of sensation in skin lesions, hands, or feet
- Aches and pains in the face or limbs
- Numbness, "sleepy" or "dead" feelings
- Paresthesias described as "ants running under the skin" in affected areas

Skin lesions typically present as hypopigmented or erythematous macules, papules, nodules, or plaques, which may be skin-colored or slightly red.

Clinically, leprosy is diagnosed when a patient exhibits at least two of the three cardinal signs. However, in endemic areas, the WHO considers the presence of a single cardinal sign sufficient for diagnosis—though this approach sometimes overlooks nerve enlargement.

The three well-established cardinal signs of leprosy (recognized for over 100 years) are:

1. **Loss of sensation in a skin lesion**
2. **Enlarged peripheral nerve**
3. **Positive skin smears**

Currently, there are no specific laboratory tests to confirm leprosy. Available tests support classification, monitoring, and detection of relapse but are not diagnostic on their own.

Sensory loss is assessed using a wisp of cotton wool. The test involves gentle touch (not brushing) of the lesion and surrounding skin to evaluate perception.